

relation. Similarly, for (iii) just above, the number 5 does not divide 9, nor 8 divides 5. Hence, 5 and 9 are not related to each other under divisibility relation.

A. Partially ordered set (Poset): An ordered pair (X, R) – where X is non-empty set and R , a partial order relation on X – is called **a partially ordered set** (for short, **poset**). For example, $(\mathbb{N}, \text{is-divisor-of})$ is a partially ordered set.

B. Total order: For a partial order R on a set X , if for every pair $x, y \in X$, either $(x, y) \in R$, or $(y, x) \in R$, i. e. every pair of elements is related, then the relation R is called **Total order**, and (X, R) is called a **totally ordered set**. For example, If for the relation *is-divisor-of* is defined on the set $\{2^0=1, 2^1, 2^2, 2^3, \dots\}$, of all powers of 2, then *is-divisor-of* is a total order on the set. In general, the relation *is-divisor-of* defined on the set $\{p^0=1, p^1, p^2, p^3, \dots\}$, of all powers of p , for a prime number p , is a total order.

C. Hasse Diagram

In partial order theory, Hasse Diagram is a graphical representation of a finite partially ordered set obtained as follows. Let (X, R) be a finite partially ordered set. The Hasse diagram is obtained by drawing *an upward line from x to y* whenever for $x, y \in X$, we have $x R y$, (i.e., $(x, y) \in R$) as shown and explained in the following examples.

Example: Let $X = \text{divisors of } 12 = \{1, 2, 3, 4, 6, 12\}$, and for $x R y$ if x is a proper divisor of y , for $x, y \in R$. Then the corresponding Hasse diagram is shown in Fig. 3.5.

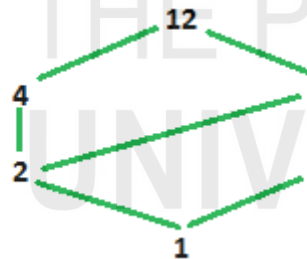


Fig. 3.5: Hasse Diagram 1

Example: Let $X = \text{Power set of } \{x, y, z\}$, i. e. $X = \{\emptyset, \{x\}, \{y\}, \{z\}, \{x, y\}, \{x, z\}, \{y, z\}, \{x, y, z\}\}$. Define R on X such that for $A, B \in X$, then $(A, B) \in R$, if $A \subset B$. The Hasse diagram is as shown in Fig 3.6.

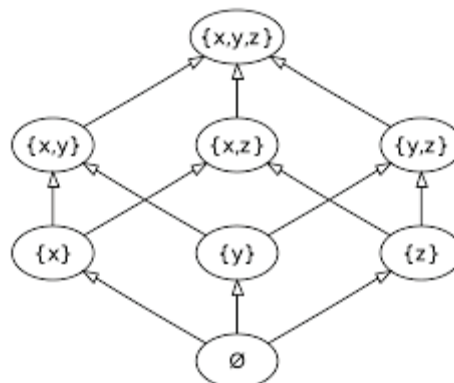


Fig. 3.6: Hasse Diagram 2

Example: Let $X = \{3, 4, 12, 24, 48, 72\}$, and under the relation R , for $x, y \in X$, $(x, y) \in R$, if x is proper divisor of y . The Hasse diagram is shown in Fig 3.7.

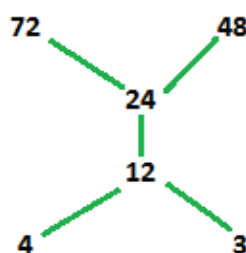


Fig. 3.7: Hasse Diagram 3

D. Some Definitions

Here we provide definitions of maximal element, minimal element, maximum/ greatest element, minimum/ least element of a set under a partial order.

In the previous Fig. 3.7 there are two elements, viz. 4 and 3, at the lowest level in the sense that $x \in X$ does not divide 3, for $x \neq 3$, and $x \in X$ does not divide 4 for $x \neq 4$. Such elements are called **minimal elements**.

Thus, in a poset (X, R) , an element a of X is a **minimal element** under R , if there does not exist x in X , such that x is related to a . A poset may have, one, two or more minimal elements. We have also shown in previous example that we also have cases of **two minimal elements**. In cases when we just have one minimal element, we also call it **minimum/least element**.

Maximal Element: In Figure 3.7, there are two elements, viz. 48 and 72, at the highest level in the sense that 48 does not divide $x \in X$, for $x \neq 48$, and also 72 does not divide $x \in X$ for $x \neq 72$. Such elements are called **Maximal elements**.

Thus, in a poset (X, R) , an element a of X is a **maximal element** under R , if there does not exist y in X , such that a is related to y . A poset may have, one, two or more **maximal elements**. In the example corresponding to Fig.3.7, we have two maximal elements, viz. 48 and 72.

In the example corresponding to Fig.3.5 we have **just one** maximal element, viz. 12. In such cases, the maximal element is also called **maximum/greatest element of the poset**.

3.2.5 Operations on Relations

A **relation** on input of required number of arguments returns true/ false, e. g. *is-less-than* $(3, 5)$ is true, and *is-less-than* $(5, 3)$ is false. On the other hand, an **operation** on input of required number of arguments returns a new element of the domain, e. g. *Sum* $(3, 5)$ returns 8.

In this section, we discuss **operations on relations**, i. e., operations for which inputs are relations, and the output is again a relation. Unless otherwise mentioned, all operations are **binary relations on** (a single) set, say, X .

The operations which we discuss may be classified as

- a) **Set-type operations:** union, intersection and difference of relations
- b) **Mapping-type operations:** inverse of operation, composition
- c) **Closure operations:** reflexive-closure, symmetric-closure, transitive closure.

A. Set-type operations on Relations: The set-type operations on relations include union of relations, intersection of relations etc. *The definitions of these operations are exactly on the lines of the definitions of operations on sets given in section 1.2.7.2 of Unit 1.* We explain this type of operations on relations through examples only. For the discussion of this type of operations on relations, let R and S be two relations **from** $X = \{1, 2, 3, 4\}$ **to** $Y = \{a, b, c\}$, with

$$R = \{(1, a), (2, a), (2, c), (3, b), (4, b), (4, c)\}$$

$$S = \{(1, b), (2, a), (3, a), (3, b), (4, a)\}$$

Then, **Universal relation U** has $4 \times 3 = 12$ elements of the form (x, y) with all $x \in X$, and all $y \in Y$.

- i) **Union** of R and S denoted by $R \cup S$, is given by

$$R \cup S = \{(1, a), (2, a), (2, c), (3, b), (4, b), (4, c), (1, b), (3, a), (4, a)\}$$

- ii) **Intersection** of R and S denoted by $R \cap S$, is given by $R \cap S = \{(2, a), (3, b)\}$

- iii) **Difference** $R - S = \{(1, a), (2, c), (4, b), (4, c)\}$, and

$$\text{Difference } S - R = \{(1, b), (3, a), (4, a)\}$$

- iv) **Complement** of relation $R = R' = \{(1, b), (1, c), (2, b), (3, a), (3, c), (4, a)\}$.

B. Mapping-type operations: inverse of operation, composition.

- i) For a relation R from set X to Y, the **inverse relation** of R, denoted by R^{-1} is the relation from Y to X, such that $(y, x) \in R^{-1}$, if and only if $(x, y) \in R$. For example, if $R = \{(1, a), (2, a), (2, c), (3, b), (4, b), (4, c)\}$ **from** $X = \{1, 2, 3, 4\}$ **to** $Y = \{a, b, c\}$ then $R^{-1} = \{(a, 1), (a, 2), (c, 2), (b, 3), (b, 4), (c, 4)\}$ is **from** $Y = \{a, b, c\}$ **to** $X = \{1, 2, 3, 4\}$. *Note the difference between R' and R^{-1} .*

- ii) **Composition of relations** denoted by $R \circ S$.

Here, R and S are defined in a slightly different way than these were defined just above.

Let R be a relation **from** $X = \{1, 2, 3, 4\}$ **to** $Y = \{a, b, c\}$, say, $R = \{(1, a), (2, a), (2, c), (4, c)\}$;

And, S be a relation **from** $Y = \{a, b, c\}$ **to** $Z = \{\text{triangle}, \text{square}\}$, say $S = \{(a, \text{triangle}), (b, \text{square}), (b, \text{triangle}), (c, \text{square})\}$. **Note the Codomain of $R = \text{Domain of } S$, is essential for the definition of composition.**

Then the relation $R \circ S$ is **from** $X = \{1, 2, 3, 4\}$ **to** $Z = \{\text{triangle}, \text{square}\}$, given by

$$R \circ S = \{(1, \text{triangle}), (2, \text{triangle}), (2, \text{square}), (4, \text{square})\}$$

C. Closure operations: reflexive-closure, symmetric-closure, transitive closure. These operations are relevant only for a **relation R on** a set, say X . The operations are illustrated through examples:

Let $X = \{1, 2, 3\}$, and $R = \{(2, 2), (3, 3), (2, 3), (3, 1)\}$

- i) **Reflexive-closure of a relation R** is a relation S such that $R \subseteq S$, and S is obtained from R by adding minimum number of ordered pairs (x, x) so that S is reflexive. For example, for $R = \{(2, 2), (3, 3), (2, 3), (3, 1)\}$, Reflexive-closure $(R) = \{(2, 2), (3, 3), (2, 3), (3, 1), (1, 1)\}$
- ii) **Symmetric-closure of a relation R** is a relation S such that $R \subseteq S$, and S is obtained from R by adding minimum number of ordered pairs (x, y) so that S is symmetric. For example, for $R = \{(2, 2), (3, 3), (2, 3), (3, 1)\}$, Symmetric-closure $(R) = \{(2, 2), (3, 3), (2, 3), (3, 1), (3, 2), (1, 3)\}$
- iii) **Transitive-closure of a relation R** is a relation S such that $R \subseteq S$, and S is obtained from R by adding minimum number of ordered pairs (x, y) so that S is transitive. For example, for $R = \{(2, 2), (3, 3), (2, 3), (3, 1)\}$, Transitive-closure $(R) = \{(2, 2), (3, 3), (2, 3), (3, 1), (2, 1)\}$.

Check Your Progress 5

1) Under the relation of *is-proper-divisor-of*, draw Hasse diagram for set X , where $X =$ all divisors of

- (i) 30 (ii) 24 (iii) 60 (iv) 32

and for each case, write all the minimal elements, and maximal elements.

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3.3 FUNCTION

Now we introduce another fundamental concept of Mathematics, viz. that of a *function*, which is a *special case of relation*, but *more general than that of 'operation'* on a set.

3.3.1 Conceptualising Function

A function is a *mathematical* term, which is a tool used to describe some aspects of the *real* world, including the following:

- a) Given that a vehicle moves at a constant average speed (say 50 km./hour), then the *distance* (say d) covered depends on the *time* (say $t \text{ hours}$) over which the vehicle moves,
- b) If a person takes a loan L (say, of Rs. 50,000/-) on simple interest R (say, at 6% per annum), the amount (say A) payable depends on the time T , after which the payment is made, *in addition to L and R* .
- c) The area (say *Area*) of a circle depends on the radius (say r) of the circle.

The statements (a), (b) and (c) above express some *real* life situations, and each of which can be *mathematically* expressed respectively as (a) $d = 50 * t$, (a) $A = 50000 (1+6/100)^t$, and $\text{Area} = \pi.r^2$.

In each of these cases, there are two *variable quantities* involved: one on LHS and the other on RHS. For example, in (a) the quantity d occurs on LHS and quantity t occurs on RHS; in (b) the quantity A occurs on LHS and quantity t occurs on RHS; and in (c) the quantity *Area* occurs on LHS and quantity r occurs on RHS.

For further discussion, first, we consider only (a) above. The value of d , occurring on L.H.S., is given by RHS, viz. $50*t$, and $50*t$ depends on the value of t *only*. For example, if $t = 3(\text{hours})$, the distance covered, i. e. $d=50* t =50*3=150 \text{ km.}$; if $t= 5 \text{ hours}$, then distance covered, i. e. $d = 50*5=250 \text{ km.}$ Further, the dependence of $50*t$ on t is *unique* in the sense that for $t=3$, we get the *unique* value 150 or $50*3$, and for $t=5$, we get the *unique* value 250 or $50*5$. This *unique dependence* of $50*t$, and hence of variable quantity d is, in mathematical terms, called **functional** dependence, and $50*t$ is denoted as **$f(t)$** , for *some* f , and t inside shows that it depends on variable quantity t .

In view of the discussion above, f is called the **function name**, the variable quantity t is called the **independent variable** and the variable quantity d is called the **dependent variable**, and the relation between d and t is called **d 's functional dependence on t** .

Going back briefly to (b), $A = 50000 (1+6/100)^t$. Let $g(t) = 50000 (1+6/100)^t$, then $A = g(t)$. Here, g is a function name, t is independent variable, and A is the

dependent variable, *and the relation between A and t is called A's functional dependence on t.*

In case of (c), let $h(r) = \pi \cdot r^2$, then $\text{Area} = h(r)$, and h is a function name, r is the independent variable, & Area is the dependent variable, *and the relation between Area and r is called Area's functional dependence on r.*

3.3.2 Function: Definition and Examples

A function from a set, say A , to a set, say B , is a special type of relation where (i) each element $a \in A$ is in relationship with some element $b \in B$, and (ii) there is exactly one element $b \in B$ in relationship with each element $a \in A$. More explicitly,

For two given *non-empty* sets X and Y , *a function f from X to Y* , denoted as

$$f: X \rightarrow Y$$

is a *rule* which associates to *each* element x of X a *unique* element, say, y of Y . The unique element y associated with x of X is denoted as $f(x)$, i. e. $y=f(x)$.

For the function $f: X \rightarrow Y$, the set X is called the **domain** of f and Y is called the **codomain** of f . Also, if $f(x)=y$, then y is called **image** of x under f , and x is called **pre-image** of y . Also, the y , or $f(x)$, is called the *value* of x under the function f . The names '**map**' and '**mapping**' are also used instead of the name '**function**'. The subset of Y which consists of all images $f(x)$ of elements of X is called the **range** of function f (in Y)

Remarks:

- *The role/ meaning of ' $\rightarrow$ ' here is not at all logical as discussed earlier. It is just a part of notation for a function. It just indicates that the symbol X , immediately on left of ' $\rightarrow$ ' is the *domain*, and the symbol Y , immediately on the right of ' $\rightarrow$ ' is the *codomain* of the function-symbol f , the symbol which precedes ' $\cdot$ '.*
- Also, in our discussion so far, in each case there is only ONE independent variable associated with each dependent variable. However, for a dependent variable, there may be **more than one independent variable**. For example, **A-Rect**, the area of a rectangle of length l and breadth b is given by $A\text{-Rect} = l*b$. The dependent variable $A\text{-Rect}$ depends on two independent variables, viz. l and b . It may be written as $A\text{-Rect} = f_1(l, b)$, where $f_1(l, b) = l*b$. Note the comma in $A\text{-Rect} = f_1(l, b)$. Similarly, the volume v of a railway container of length l , breadth b and height h is given by $v = l*b*h$. Then the volume v may be written as a function $f_2(l, b, h)$ of three independent variables l , b , and h .

However, in the rest of the discussion, we will discuss functions only of **one independent variable with domain R** , the set of real numbers; and hence, it will be of the form $f: R \rightarrow Y$, where R is the set of real numbers. A function having its

domain as R or its subset is called a **real function**. We may note that if $g: X \rightarrow Y$ is a function, and if $Y=R$, i. e. $g: X \rightarrow R$, then g is called a **real-valued function**. In the next subsection, we discuss some specific single-variable functions that are frequently used in practical applications, especially in the field of economics.

As mentioned just above, if $Y=R$, the function $f: X \rightarrow R$ is called **real-valued** function. If $Y=C$, then $f: X \rightarrow C$ is called **complex-valued** function. Similarly, by taking Y , the codomain, as Z or Q , we get respectively **integer-valued** function and **rational-valued** functions.

Let us look at some more examples for illustrating the concept of function.

i) Let $X=Y=N$, the set of natural numbers, then

$$f: N \rightarrow N,$$

$$\text{such that } f(x) = 2x+1, \forall x \in N.$$

Clearly, for each $x \in N$, $2x+1$ exists in N and is unique for each x . Hence, $f: N \rightarrow N$ is a function.

ii) Let $f: Z \rightarrow N$, such that $f(x) = x^2 + 3$,

then for each $x \in Z$, $x^2 + 3 \in N$ and is unique, therefore, f is a function

iii) Let $f: Z \rightarrow N$, such that $f(x) = x^2 - 3$,

then for $x=0$, $x^2 - 3 = -3 \notin N$, Hence, $f: Z \rightarrow N$ is *not* a function from Z to N .

iv) Let $f: Z \rightarrow Z$, such that $f(x) = x^2 - 3 \in Z$

then $x^2 - 3 \in Z$ (codomain), $\forall x \in Z$ (domain). Hence, $f: Z \rightarrow Z$ is a function from Z to Z .

(Note: Change of codomain has converted a non-function to a function)

3.3.3 Operations as Functions

The *binary operation* '+' on Natural numbers may be considered the *function*

Plus: $N \times N \rightarrow N$, such that

Plus $(x, y) = (x + y)$, for each pair of Natural numbers.

The *union operation* $X \cup Y$ on sets can be considered as a *function* **Union:** $\mathcal{P} \times \mathcal{P} \rightarrow \mathcal{P}$,

where $\mathcal{P}$ is a set of all subsets of universal set U , such that **Union** $(X, Y) = X \cup Y$, for each pair X, Y of sets from $\mathcal{P}$.

In general, an operation is a function in which the domain of function is either the codomain or cross-product of codomain of function, Cross-product may involve the codomain finitely many times: Codomain $\times$ Codomain $\times \dots \times$ Codomain. For example, the *square operation* on N is the function $SQ: N \rightarrow N$, such that $SQ(n) = n^2$. The domain of function SQ is the codomain of the

function SQ . In the case of Plus and Union, just above, the Domain is a cross-product of codomain.

3.3.4 Types of Functions

By definition of function $f: X \rightarrow Y$, for each x , its image $f(x)$ must belong to Y and $f(x)$ should not have more than one value in Y , as happens in the case of $f(x) = \text{square-root of } x$. **However, there is no restriction on elements of X , for f to be a function.** For example, for the function Mod, and for both values 5 & -5 , $\text{Mod}(5) = \text{Mod}(-5) = 5$.

3.3.4.1 Injective function

A function $f: X \rightarrow Y$ is called **injective**, if for x_1, x_2 in X with $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$.

The definition can be equivalently stated as:

A function $f: X \rightarrow Y$ is called **injective**, if $f(x_1) = f(x_2)$ then $x_1 = x_2$

Note: The term **1-1** is also used instead of **injective**. However, there is a similar term **1-to-1** used for a distinct concept viz. **bijective**. **For this reason, the terms 1-1 and 1-to-1 may be avoided.**

Examples:

i) The function

$$f: \mathbb{N} \rightarrow \mathbb{N},$$

such that $f(x) = 2x+1, \forall x \in \mathbb{N}$ is injective. Let $f(x) = f(y)$, then $2x+1 = 2y+1$ implying $x=y$

ii) The function

$$f: \mathbb{N} \rightarrow \mathbb{R}, \text{ defined by the rule}$$

$f(x) = \text{positive square-root of } x$ is injective. Let $f(x) = f(y)$, i. e. $\sqrt{x} = \sqrt{y}$ squaring both sides, we get $x = y$.

iii) The function

$$f: \mathbb{N} \rightarrow \mathbb{R}, \text{ defined by the rule}$$

$f(x) = x^2$ is injective, as, for x, y from $\mathbb{N}$, if $f(x) = f(y)$, i. e. $x^2 = y^2$ then $x = y$ or $x = -y$, but as y is in $\mathbb{N}$, the number $-y$ cannot be in $\mathbb{N}$. Hence, $x = y$.

Procedure (generally adopted) to establish a function as injective:

For function $f: X \rightarrow Y$,

- i) assume that $f(x) = f(y)$ and then show $x = y$ (must) for $x, y \in X$ or alternatively, in order to establish **f is NOT injective**, show for **some** pair of elements x, y , with $x \neq y$ of X , that $f(x) = f(y)$.

The latter method was used in showing $f: \mathbf{Z} \rightarrow \mathbf{R}$, defined by the rule $f(x) = x^2$, is *not* injective by using $f(2) = f(-2) = 4$. ***We have already used the first one to show*** the function $f: \mathbf{N} \rightarrow \mathbf{R}$, defined by the rule $f(x) = x^2$ as injective.

3.3.4.2 Surjective/ Onto Function

For a function $f: X \rightarrow Y$, it is essential that *every element of X (domain)* corresponds to some element of Y . But, for f to be a function, it is *not* compulsory that for every y of Y (*codomain*), there be a corresponding element x of X such that $f(x) = y$.

For example, for the function $f: \mathbf{N} \rightarrow \mathbf{N}$, with $f(x) = 2x+1 \quad \forall x \in \mathbf{N}$,

For the element 4 of $\mathbf{N}$ (the codomain) *an x in $\mathbf{N}$ (the domain)*, if exists, satisfies $2x+1=4$, i. e. $x = 3/2$, which is not in $\mathbf{N}$. ***Hence f is not surjective.***

A function $f: X \rightarrow Y$ is called **surjective**, if for an arbitrary element y of Y (*the codomain*), we are able to find an x in X (*the domain*) such that $f(x) = y$.

Procedure for establishing a function as surjective/onto

Take an *arbitrary* element y of the *codomain* Y , and show the existence of some element x of domain X with $f(x) = y$. If it is possible to find such an element x of *domain* X , then the function is surjective, *else*, it is *not* surjective.

Example: For the function $\text{Mod}: \mathbf{Z} \rightarrow \mathbf{N}$ such that

$$\text{Mod}(x) = |x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$$

let x be in the codomain $\mathbf{N}$, then this x belongs to $\mathbf{Z}$, the domain (*because $\mathbf{N} \subseteq \mathbf{Z}$*) with $|x| = x$, hence, Mod is surjective.

Next, consider $f: \mathbf{Z} \rightarrow \mathbf{C}$, defined by the rule $f(x) = \text{positive square-root of } x$.

Let, for x in $\mathbf{Z}$, with $f(x) = 3.75$, i. e. positive-square-root of $x = 3.75$. Therefore, $x = (3.75)^2$, which is not in $\mathbf{Z}$. Hence, ***$f: \mathbf{Z} \rightarrow \mathbf{C}$ is not surjective***

However, we can show that **$f: \mathbf{C} \rightarrow \mathbf{C}$** , the set of complex numbers, defined by the rule $f(x) = \text{positive square-root of } x$, is surjective.

3.3.4.3 Bijective Function/ 1-to-1 correspondence

A function $f: X \rightarrow Y$ is called **Bijective** if (and only if) it is both injective and surjective. A bijective function from X to Y is also called **1-to-1 correspondence** between X and Y .

Check Your Progress 6

1) Find the domain and range of the following *real-valued real* functions:

(i) $f(x) = -|x|$ (ii) $f(x) = +\sqrt{9-x^2}$

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- 2) A real-valued real function f is defined by $f(x) = 2x - 5$. Write down the values of (i) $f(0)$, (ii) $f(7)$, (iii) $f(-3)$.

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- 3) For each of the following, tell whether f is a function or not. Justify your answer.

- i) $f: \mathbb{N} \rightarrow \mathbb{N}$, such that $f(x) = x - 3$ (ii) $f: \mathbb{N} \rightarrow \mathbb{Z}$, such that $f(x) = x - 3$

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3.4 LET US SUM UP

Relation and Function are two of the most fundamental concepts of Mathematics. In this unit, we introduced the concepts through respective definitions and illustrative examples. Also, we discussed three main properties of relation, viz. reflexivity, symmetry and transitivity, along with some of the variants of these properties, and some interesting combinations of these properties, viz. equivalence relation and partial order relations. Next, we discussed some operations on relations. Finally, we discussed some special type of functions, viz. injective, surjective and bijective functions.

3.5 KEY WORDS

Equivalence Relation : A relation R on a set X is said to be an **equivalence relation**, if R is reflexive, symmetric and transitive. In more detailed form, R is an equivalence relation on set X , if and only if

- R is reflexive on X , i. e. for **every** $x \in X$, we have $(x, x) \in R$, and
- R is symmetric on X , i. e. for $x, y \in X$, **if** $(x, y) \in R$, **then** $(y, x) \in R$, and
- R is transitive on X , i. e. for $x, y, z \in X$, **if** $(x, y) \in R$, **and** $(y, z) \in R$, **then** $(x, z) \in R$.

Equivalence Class of an equivalence relation : For an equivalence relation R on a set X , the subset $[a]$ of X defined as $[a] = \{x: (x, a) \in R\}$ is called an equivalence class of a under the equivalence relation R

Function : For two given *non-empty* sets X and Y , a *function* f from X to Y , denoted as $f: X \rightarrow Y$ is a *rule* which